

Cambridge IGCSE™

PHYSICS**0625/63**

Paper 6 Alternative to Practical

May/June 2026

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **13** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

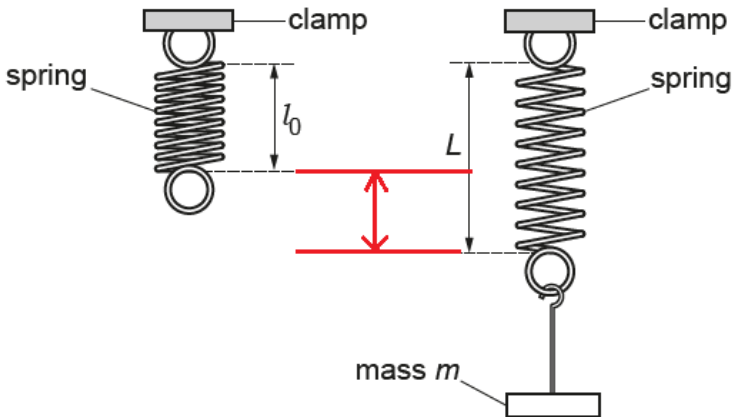
Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	arithmetic error
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response has not answered question

Annotation	Meaning
RE	rounding error
SEEN or 	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
TE	transcription error
TV	response is too vague or there is insufficient detail in response
T	answer outside the tolerance of the mark scheme
	used to highlight parts of an extended response
	used to highlight parts of an extended response
MO	mandatory mark not awarded
SC	special case
	unclear response
PD	poor diagram
POT	power of ten error
XP	incorrect physics
U	incorrect unit

Acronym / shorthand	Explanation
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word/phrase/unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
ORA	Or reverse argument
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, <u>not</u> treated as contradictory.
cao	correct answer only

Question	Answer	Marks	Guidance
1(a)(i)	$(m_1 =) 8.9 \text{ (g)}$	1	cao
1(a)(ii)	(initial reading =) $11(.0) \text{ (cm}^3\text{)}$ and (final reading =) $17(.0) \text{ (cm}^3\text{)}$	1	cao
1(a)(iii)	$(V_1 =) 1.2 \text{ (cm}^3\text{)}$	1	ecf 1(a)(ii) , any significant figures > 1, ignore rounding
1(a)(iv)	view scale perpendicularly / view bottom of meniscus	1	accept view scale at eye level
1(a)(v)	$(\rho_1 =) 7.4$	1	ecf 1(a)(i) and 1(a)(ii) , accept use of 8.89 for m_1 any significant figures > 1, ignore rounding
	g / cm^3	1	accept correct conversion to kg / m^3
1(b)(i)	$(h_5 =) 1.4 \text{ (cm)}$	1	accept 14 mm do not accept 1.40 (cm) or 14.0 (mm)
1(b)(ii)	$6.3 \text{ (cm}^3\text{)}$	1	ecf 1(b)(ii) , any significant figures > 1, correctly rounded
1(b)(iii)	$(\rho_2 =) 6.9 \text{ (g / cm}^3\text{)}$	1	ecf 1(b)(ii) , any significant figures > 1 ignore unit, ignore rounding accept correct conversion to kg / m^3 if unit is stated
1(c)	statement to match candidate's values of ρ_1 and ρ_2	1	10% or less must state YES > 10% must state NO
	values used in a calculation to justify the statement	1	e.g., difference calculated, and correct percentage calculation / ratio of values shown to be > 0.9 / 90% (YES) or < 0.9 / 90% (NO) if both values are exactly the same, it is sufficient to say 'the values are the same' with no calculation required, for 2 marks

Question	Answer	Marks	Guidance
2(a)(i)		1	mark by eye
2(a)(ii)	any two from: view perpendicularly measure with the ruler close / parallel to the spring use of set-square or a fiducial marker described	2	accept view at eye level ignore wait until the spring is stationary
2(b)(i)	1.8 (cm)	1	cao do not accept 1.80
2(b)(ii)	suitable linear scales (plots occupying at least $\frac{1}{2}$ grid)	1	accept ecf for any subsequent marks not awkward scales e.g. 3 s or 7 s.
	plots all correct to $\pm \frac{1}{2}$ small square <u>and</u> precise plots	1	accept diameter of plots $\leq 1/2$ square
	well-judged best-fit line <u>and</u> thin, continuous line	1	accept a straight line only accept line joining origin and last point if points are plotted correctly

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Question	Answer	Marks	Guidance
2(b)(iii)	method clearly shown on graph	1	
	extension correctly read to $\pm \frac{1}{2}$ small square	1	mark can be awarded if candidate's $L = (\text{extension inferred from graph}) + 1.9$
	$2.4 \leq L \leq 2.5$	1	accuracy mark
2(c)	any one from: wear goggles don't overload the spring (so that it breaks) secure stand to the bench (e.g. clamp)	1	ignore keep fingers / feet from being underneath the spring accept ideas about tightening the clamp

Question	Answer	Marks	Guidance
3(a)	correct symbol for voltmeter connected in parallel with the resistor	1	accept voltmeter connected across supply ignore small gaps ignore symbol for junction of conductors do not accept line through voltmeter
	correct symbol for ammeter connected in gap in circuit	1	accept ammeter anywhere in series if the gap is closed ignore small gaps
3(b)	0.3(0) (A)	1	cao
3(c)(i)	V, A, Ω	1	accept units written in words
3(c)(ii)	2.5 (Ω) or 2.50 (Ω)	1	accept 2 or 3 significant figures only, correctly rounded ecf 3(b)
3(c)(iii)	as the number of resistors (in parallel) increases, the lower the resistance / ORA	1	accept number of resistors is inversely proportional to the resistance
	use of data from Table 3.1 to justify the conclusion	1	ignore negative correlation
3(c)(iv)	(add switch to) turn circuit off between readings	1	
	use lower current / supply p.d.	1	accept use resistors of a higher value to reduce current
3(d)	0.58	1	
	appropriate reason e.g. transposed digits (when recorded), resistor disconnected, only 2 resistors connected in parallel	1	accept it should be 0.85 accept temperature increased (so resistance increased and current decreased) ignore misread ammeter

Question	Answer	Marks	Guidance
4	MP1 variable chosen valid variable which may affect the rate at which the temperature of the water increases		variable must be <u>stated</u> somewhere in the plan and not just implied in the method, table or analysis. e.g. size / surface area / colour / material of beaker e.g. volume / initial temperature of water ignore lid or no lid or salt concentration or type of water e.g. distilled / pure, or power / current / voltage / temperature of heater other marking points can be scored even if incorrect independent variable chosen
	MP2 additional apparatus thermometer and stop-watch / timer	1	can be mentioned anywhere accept clock or watch
	MP3 method measure: • initial temperature • final temperature • time and repeat for new value of independent variable	1	both points needed for the mark accept e.g. measure temperature every 30 s
	MP4 control variables any significant variable appropriate to the independent variable	1	e.g. size / surface area / colour / material of beaker e.g. volume / initial temperature of water e.g. time of heating water ignore room temperature

Question	Answer	Marks	Guidance
4	MP5 table columns, <u>with quantities and correct units</u>, for independent variable and dependent variable	1	e.g. name of independent variable and (initial temperature and) final temperature or time (to reach a certain temperature) accept plausible symbols instead of words accept units given in the body of the table accept headed rows instead of columns ignore extra columns in table, e.g. (rate of) temperature change accept h, min, s (syllabus page 50) not m or mins or sec or secs or C°
	MP6 conclusion compare readings in the table to see if / how a change in the variable produces change in rate of increase in temperature OR plot a (line) graph <u>with axes specified</u>	1	ignore a conclusion or a prediction from theory compare readings in the table, and nothing else added, is insufficient accept a set of labelled axes drawn – no units needed accept temperature change (in certain time) or rate of temperature change or time (to reach a certain temperature) accept graph of temperature against time for each value / category of the independent variable ignore a bar chart <u>unless</u> the colour / material of the beaker is the independent variable
	MP7 additional point (one from) second valid control variable stated at least five sets of data taken repeat each <u>measurement</u> and take the average additional apparatus (if needed) to measure the independent variable method of calculating rate of temperature change	1	accept this seen in the table ignore repeat the experiment (and take an average)